

Synthesis Challenge

Single-step retrosynthesis: template-free seq2seq, ensembling & reranking

Simon Danninger

JKU Linz – AI in Life Sciences

1. The challenge

Task. Single-step retrosynthesis. Given a **product** SMILES, predict the **reactant(s)**.
Training pairs are reaction SMILES reactants » products.

Data.

- ▶ data_train.csv – 40,000 reaction rows.
- ▶ product_smiles_test.csv – 10,000 products to solve.

Metric: top-1 exact match. Correct iff the set of RDKit-canonical predicted reactants equals the set of true reactants.

2. EDA

Clean & small

- ▶ 40k/40k train, 10k/10k test parse under RDKit
- ▶ Tiny molecules: product token len mean 13, max 25
- ▶ Shared vocab = 50 tokens, **0** <unk>
- ▶ Fragments per reaction: 1→62%, 2→34%, 3+→4%

Noisy & leaky

- ▶ 21.7% of test products appear in train
- ▶ But product→reactant map is **one-to-many**
- ▶ ⇒ exact-lookup precision only ~42%

3. Our approach

Encoder-decoder Transformer (product $\rightarrow$ reactants), wrapped in retrieval fallbacks and round-trip reranking.

1. **Retrieval baselines:** exact-match lookup, kNN (ECFP4 Tanimoto).
2. **Seq2seq Transformer** SMILES augmentation (randomize source on the fly), Beam search + length penalty.
3. **Forward model + round-trip rerank:** a reactants $\rightarrow$ product model scores whether a candidate reproduces the query.
4. **Hybrid cascade:** lookup $\rightarrow$ neural beam $\rightarrow$ kNN $\rightarrow$ identity, guaranteeing one valid line per row.

local validation. 2,000-row held-out split (seed 1234)

4. Versions

ver.	what	local	server (public)
v0	exact-lookup / kNN	0.114	–
v1	seq2seq Transformer (no aug, beam-5)	0.277	–
v2	+ SMILES augmentation, beam-10, TTA $\times$ 5	0.295	–
v3	+ forward model + round-trip cascade	0.299	–
v4	candidate ensemble (v1 $\cup$ v2) + cascade	0.307	0.301
v5	forward-likelihood reranking	0.328 [†]	–
v6	5-model ensemble + stronger forward + blend	0.374 [†]	0.351
v6big	bigger retro model (25 M, beam-20)	0.318	–
v7	learned reranker (logreg, 5-fold OOF)	0.366	0.352

[†] Optimistic: tuned on the reported holdout.

Two phases: **v0–v4** = build the generator; **v5–v7** = exploit reranking over a candidate pool.

5. Phase 1: building the generator

What mattered

- ▶ Lookup baseline caps at **0.114** (1-to-many noise)
- ▶ Neural seq2seq reaches **0.277**
- ▶ **Augmentation** better and prevents overfit
- ▶ Round-trip cascade + ensembling: 0.295 → **0.307**

step	local top-1
exact lookup	0.114
seq2seq (no aug)	0.277
+ augmentation	0.295
+ round-trip	0.299
ensemble+cascade	0.307

Server-confirmed: **0.301**.

6. Phase 2: the bottleneck is ranking, not coverage

Key realization. Over a 5-model candidate pool:

pool oracle = **0.82** but realized top-1 ≈ 0.45

The correct reactant set is in the pool 82% of the time, but we rank it #1 too rarely.

So we focus on the ranker.

- ▶ v5 – forward-likelihood blend
- ▶ v6 – 5 retro models + a stronger forward model + tuned blend
- ▶ v7 – **learned reranker**: logreg over candidate features (out-of-fold, no leakage)

7. Error analysis

Decomposition of the gap (2k holdout).

gold <i>not</i> in pool (coverage ceiling)	18.0%
gold in pool & picked #1	36.6%
gold in pool but mis-ranked	45.4%

- ▶ Half the mis-ranks are near-misses (**gold at rank 2: 31.6%**, rank 3: 18.9%).
- ▶ ~57% of our wrong picks are legitimate alternative reactant sets → hard to fix.
- ▶ Multi-fragment is harder: 1-frag 39% > 2-frag 34% > 3-frag 13%.

8. Final results

Server submissions (3 used).

#	version	server top-1 (public)
1	v4 ensemble + cascade	0.301
2	v6 ensemble + tuned blend	0.351
3	v7 learned reranker	0.352

Best server result: 0.352 (v6 statistically tied).

Honest reading.

- ▶ The generative-ensemble + reranking approach plateaus at ~ 0.35 : v5, v6, v7 all land there.
- ▶ v7 is preferred over v6 because its out-of-fold score is honest, while v6 overfitted on the hold-out set.
- ▶ Local validation becomes optimistic with multiple tuning iterations $\rightarrow$ always evaluate rerankers out-of-fold.

9. Lessons

1. **Diagnose before you optimize.** The pool-oracle / realized-top-1 gap (0.82 vs 0.45) shifted the problem from “generate better” to “rank better”
2. **Augmentation > architecture.** On-the-fly SMILES randomization was the single biggest training benefit and the prerequisite for scaling (robustness).
3. **Evaluate out-of-fold.** Hand-tuned blends looked great until an honest OOF reranker exposed the optimism; the local proxy drifted $+0.006 \rightarrow +0.022$ as tuned pieces stacked.

Thank you.